Milestone Test-07

KPM Dropper NEET **Answer Key & Solutions**

Physics (1–45) • Chemistry (46–90) • Botany + Zoology (91–180)

Answer Key

PHYSICS Q 1–45

1 (3)	2 (3)	3 (4)	4 (3)	5 (2)	6 (1)	7 (2)	8 (2)	9 (3)	10 (4)
11 (2)	12 (1)	13 (1)	14 (1)	15 (1)	16 (2)	17 (1)	18 (1)	19 (4)	20 (4)
21 (4)	22 (3)	23 (3)	24 (3)	25 (1)	26 (2)	27 (3)	28 (3)	29 (1)	30 (4)
31 (3)	32 (3)	33 (3)	34 (4)	35 (4)	36 (3)	37 (2)	38 (3)	39 (4)	40 (3)
41 (3)	42 (3)	43 (2)	44 (1)	45 (1)					

CHEMISTRY Q 46–90

46 (3)	47 (4)	48 (4)	49 (4)	50 (3)	51 (1)	52 (2)	53 (3)	54 (2)	55 (3)
56 (2)	57 (4)	58 (1)	59 (1)	60 (1)	61 (3)	62 (1)	63 (3)	64 (2)	65 (1)
66 (2)	67 (3)	68 (2)	69 (3)	70 (3)	71 (4)	72 (2)	73 (2)	74 (4)	75 (2)
76 (2)	77 (1)	78 (2)	79 (2)	80 (2)	81 (2)	82 (4)	83 (3)	84 (4)	85 (3)
86 (1)	87 (4)	88 (4)	89 (4)	90 (2)					

BOTANY + ZOOLOGY Q 91–180

91 (4)	92 (3)	93 (2)	94 (3)	95 (2)	96 (3)	97 (1)	98 (2)	99 (2)	100 (1)
101 (1)	102 (1)	103 (3)	104 (2)	105 (3)	106 (2)	107 (3)	108 (3)	109 (3)	110 (3)
111 (2)	112 (3)	113 (2)	114 (2)	115 (2)	116 (1)	117 (1)	118 (3)	119 (3)	120 (2)
121 (1)	122 (1)	123 (3)	124 (4)	125 (3)	126 (3)	127 (2)	128 (4)	129 (2)	130 (2)
131 (3)	132 (1)	133 (3)	134 (2)	135 (2)	136 (3)	137 (2)	138 (3)	139 (3)	140 (3)
141 (2)	142 (2)	143 (1)	144 (4)	145 (1)	146 (3)	147 (4)	148 (1)	149 (2)	150 (3)
151 (1)	152 (2)	153 (2)	154 (4)	155 (1)	156 (1)	157 (3)	158 (4)	159 (3)	160 (1)
161 (2)	162 (3)	163 (2)	164 (3)	165 (3)	166 (2)	167 (4)	168 (2)	169 (3)	170 (1)
171 (3)	172 (2)	173 (4)	174 (4)	175 (2)	176 (2)	177 (2)	178 (4)	179 (3)	180 (3)

PHYSICS (Q 1–45) Solutions

Q1. Breaking stress sirf material par depend karta hai, length/radius/shape par nahi. Ans. (3)

Q2. Freely falling lift mein $g_{\text{eff}} = 0$. $h = \dfrac{2T\cos\theta}{r\rho g} \to \infty$, to water poori tube (20 cm) bhar dega. Ans. (3)

Q3. Net force $= f - \dfrac{f}{2} = \dfrac{f}{2}$, to $a = \dfrac{f}{2m}$. Right half (mass $\frac{m}{2}$) lo: $f - T = \dfrac{m}{2}a = \dfrac{f}{4} \Rightarrow T = \dfrac{3f}{4}$. Stress $= \dfrac{3f}{4A}$. Ans. (4)

Q4. Meniscus ka radius adjust ho jata hai ($R = \dfrac{r}{\cos\theta}$). Water top tak aata hai aur overflow nahi hota. Ans. (3)

Q5. Slope = Young's modulus. T_1 ka slope zyada hai, aur Y temperature badhne par ghatta hai, to $T_1 < T_2$. Ans. (2)

Q6. Liquid tab chadhta hai jab contact angle acute ho ($\theta < 90°$). Ans. (1)

Q7. $U = \dfrac{F^2L}{2AY} \propto \dfrac{L}{d^2}$. A: $\frac{3}{4}$, B: 1, to ratio $3:4$. Ans. (2)

Q8. $\Delta l = \dfrac{\rho g L^2}{2Y} = \dfrac{1.5 \times 9.8 \times (0.08)^2}{2 \times 5 \times 10^8} \approx 9.6 \times 10^{-11}$ m. Ans. (2)

Q9. $P = P_0 + \dfrac{2T}{r} = 1.013 \times 10^5 + \dfrac{2 \times 0.07}{10^{-4}} = 1.013 \times 10^5 + 1400 \approx 1.027 \times 10^5$ Pa. Ans. (3)

Q10. $Y = \dfrac{\text{stress}}{\text{strain}} = \dfrac{4 \times 10^7}{2 \times 10^{-4}} = 2 \times 10^{11}\ \text{N m}^{-2}$. Ans. (4)

Q11. Temperature badhne par liquid ki viscosity ghatti hai. Ans. (2)

Q12. Hooke's law sirf proportional limit tak (OA) follow hota hai. Ans. (1)

Q13. $P = \dfrac{mg}{A} = \dfrac{3000 \times 9.8}{425 \times 10^{-4}} = 6.92 \times 10^5\ \text{N m}^{-2}$. Ans. (1)

Q14. Sudden loading mein stress gradual loading se do guna hota hai. Ans. (1)

Q15. Water ka density 4°C par maximum hota hai, pehle badhta hai phir ghatta hai. Graph (1) match karta hai. Ans. (1)

Q16. Shear stress ka SI unit $\text{N m}^{-2} = \text{Nm}^{-2}$. Ans. (2)

Q17. Chhoti boondon mein surface tension dominate karta hai (spherical), badi mein gravity. A aur R dono sahi hain aur R, A ki sahi explanation hai. Ans. (1)

Q18. Loading aur unloading curve ke beech ka area energy loss (heat) dikhata hai, to III sahi hai. I aur II graph se deduce nahi hote. Ans. (1)

Q19. Streamline flow mein ek fixed point par velocity constant rehti hai, kisi particle ki hamesha nahi. Isliye none. Ans. (4)

Q20. Smooth surface par string accelerate hoti hai, to tension $T = \dfrac{Fx}{L_0}$. $U = \displaystyle\int_0^{L_0} \frac{T^2\,dx}{2AY} = \frac{F^2L_0}{6AY}$. Ans. (4)

Q21. $\tan\theta = \dfrac{a}{g} = \dfrac{19.6}{9.8} = 2$, to $\sin\theta = \dfrac{2}{\sqrt{5}}$, $\theta = \sin^{-1}\left(\dfrac{2}{\sqrt{5}}\right)$. Ans. (4)

Q22. Elastomers (rubber, aorta) ka stress–strain curve non-linear hota hai (shuru mein flat, phir steep). Ans. (3)

Q23. $\Delta V = \dfrac{M}{\rho} - \dfrac{M}{\sigma} = M\left(\dfrac{1}{\rho} - \dfrac{1}{\sigma}\right)$. Ans. (3)

Q24. Radius double karne par area 4 guna, to stress $\frac{1}{4}$ ho jata hai. Statement I galat, II sahi. Ans. (3)

Q25. $F = \rho g h A = 900 \times 10 \times 0.4 \times 10^{-3} = 3.6$ N. Ans. (1)

Q26. $Y = \tan\theta$. $Y_A = \tan 60° = \sqrt{3}$, $Y_B = \tan 30° = \dfrac{1}{\sqrt{3}}$, to $Y_A = 3Y_B$. Ans. (2)

Q27. Bottom par pressure $= \rho g h$, jo bottom ke area par depend nahi karta. Ans. (3)

Q28. Compressibility = fractional volume change per unit pressure change. Ans. (3)

Q29. $F = T \times 2\pi r = 0.07 \times 2\pi \times 0.045 \approx 19.8$ mN. Ans. (1)

Q30. $Y = \dfrac{2 \times 10^6}{0.05} = 4 \times 10^7\ \text{N m}^{-2}$. Ans. (4)

Q31. Interface level par pressure balance: $4 \times 5 + x = h_2$ (x = arm 1 mein water column). Water volume: $x + h_2 = 40$, to $x = 10$, $h_2 = 30$, $h_1 = 15$. $\dfrac{h_2}{h_1} = \dfrac{2}{1}$. Ans. (3)

Q32. Hooke's law mein $\dfrac{\text{stress}}{\text{strain}} = Y$ (constant). Ans. (3)

Q33. $v = \sqrt{2gD}$, $t = \sqrt{\dfrac{2(H-D)}{g}}$, to $x = vt = 2\sqrt{D(H-D)}$. Ans. (3)

Q34. Shear stress $= \dfrac{5.6 \times 10^6}{0.04} = 1.4 \times 10^8$ Pa. Strain $= \dfrac{1.4 \times 10^8}{5.6 \times 10^9} = 0.025$. Displacement $= 0.025 \times 20$ cm $= 5$ mm. Ans. (4)

Q35. Bernoulli: spinning ball, carburettor, mixer blades mein lagta hai. Heart attack mein nahi. Ans. (4)

Q36. Breaking stress length/area par depend nahi karta, to I galat. II sahi. Ans. (3)

Q37. $\frac{1}{2}\rho v^2 = \rho_{\text{Hg}}\, g\, h \Rightarrow v^2 = 2 \times 13.6 \times 10 \times 0.36 \approx 97.9$, $v \approx 9.8$ m/s. Ans. (2)

Q38. I: stress vector nahi hai (sahi). II: ek direction ka force doosri directions mein bhi strain de sakta hai (sahi). Ans. (3)

Q39. $\Delta E = 4\pi R^2 T\left(n^{1/3} - 1\right) = 4\pi R^2 T \times 9 = 36\pi R^2 T$. Ans. (4)

Q40. Steel zyada elastic hai kyunki same force par kam deform hota hai. A, R sahi aur R explanation hai. Ans. (3)

Q41. $P_A = 10^5 + 1000 \times 10 \times 10 = 2 \times 10^5$. $P_B = 2 \times 10^5 + 2000 \times 10 \times 20 = 6 \times 10^5$. Ratio $= \dfrac{1}{3}$. Ans. (3)

Q42. $\Delta l \propto \dfrac{1}{r^2}$. Diameter half $\Rightarrow \Delta l$ 4 guna $= 4$ cm. Ans. (3)

Q43. $U = \frac{1}{2}kx^2 = \dfrac{T^2}{2k}$ (jahan $T = kx$). Ans. (2)

Q44. Energy density $= \frac{1}{2} \times$ stress $\times$ strain $= \frac{1}{2} Y\,(\text{strain})^2$. Options (1) aur (4) same likhe hain, to (1) mark kiya. Ans. (1)

Q45. Pulley ke dono sides W hone par wire mein tension W hi rehta hai, to elongation l. Ans. (1)

CHEMISTRY (Q 46–90) Solutions

Q46. Chain: $BrCH_2-CH(CH_3)-CH_2CH_3 \rightarrow$ 1-Bromo-2-methylbutane. Ans. (3)

Q47. $\Delta H = \Delta U + \Delta n_g RT$. $\Delta H < \Delta U$ tab hota hai jab $\Delta n_g < 0$ (products ke moles kam). Ans. (4)

Q48. State functions: H, S, G. Heat path function hai. Ans. (4)

Q49. Benzylic $-OH$, HBr se $-Br$ ban jata hai. Phenolic $-OH$ nahi badalta. Product m-$HO-C_6H_4-CH_2Br$. Ans. (4)

Q50. Aryl halide mein EDG ($-CH_3$) nucleophilic substitution ghatata hai. Rate: C < B < A. Ans. (3)

Q51. Ideal gas ki internal energy sirf temperature par depend karti hai. Ans. (1)

Q52. Halogen acids ke saath reactivity $3^\circ > 2^\circ > 1^\circ$, to C > B > A. Ans. (2)

Q53. $\Delta G = \Delta H - T\Delta S$. $\Delta H = +$ve aur $\Delta S = -$ve par ΔG hamesha $+$ve, to non-spontaneous. Ans. (3)

Q54. Vinyl chloride aur chlorobenzene mein C–Cl bond mein partial double bond character hai (resonance), to unreactive. Chloroethane sabse reactive. Order: vinyl chloride < chlorobenzene < chloroethane. Ans. (2)

Q55. Vacuum mein free expansion mein $P_{ext} = 0$, to $w = 0$. Ans. (3)

Q56. 3° halide S_N1 dete hain, S_N2 nahi (steric hindrance). A galat, R sahi. Ans. (2)

Q57. β-carbon par benzylic H hai, aur CH_3O^- strong base hai. E2 hota hai aur conjugated alkene (2-phenylbut-1-ene) banta hai. *(Note: yeh question thoda tricky hai; S_N2 wale option (2) ka bhi chance hai, lekin E2 zyada likely hai.)* Ans. (4)

Q58. Khule container mein chai mein matter aur energy dono exchange hote hain, open system. Ans. (1)

Q59. A sahi ($SOCl_2$ prefer hota hai kyunki by-products gas hain). R galat (phosphorus chlorides impure product dete hain). Ans. (1)

Q60. $\Delta S = \dfrac{\Delta H}{T} = \dfrac{300000}{300} = 1000\ \text{J K}^{-1}\ \text{mol}^{-1}$. Ans. (1)

Q61. Primary carbocation $\rightarrow$ ring expansion $\rightarrow$ stable 2° cyclopentyl cation $\rightarrow$ chlorocyclopentane. Ans. (3)

Q62. $q_p = \Delta H$ (sahi). $\Delta H = \Delta U$ sirf $\Delta n_g = 0$ par hota hai, "always" galat. Ans. (1)

Q63. Cyclohexyl–MgBr + $H_2O \rightarrow$ cyclohexane (B). Ans. (3)

Q64. Vacuum ke against expansion mein $w = 0$, to A galat. Work path function hai, R sahi. Ans. (2)

Q65. 3° alcohol, Lucas reagent (conc. $HCl/ZnCl_2$) se room temperature par turant 3° chloride deta hai. Ans. (1)

Q66. $w = -$ve matlab system work energy khota hai aur surroundings par kaam karta hai. Ans. (2)

Q67. S_N1 polar protic solvent mein hota hai (sahi). S_N2 mein inversion hota hai (sahi). Ans. (3)

Q68. Adiabatic mein $q = 0$, to $\Delta U = w$. Ans. (2)

Q69. Branching badhne par b.p. ghatta hai. Order: tert-butyl bromide < isobutyl bromide < n-butyl bromide, yaani (iii) < (i) < (ii). Ans. (3)

Q70. $NH_4NO_3(s) \longrightarrow N_2O(g) + 2\,H_2O(g)$: solid se gas, $\Delta n_g = +3$, to $\Delta S > 0$. Ans. (3)

Q71. S_N2 sabse tez least hindered primary iodide par hota hai: cyclohexylmethyl iodide. Ans. (4)

Q72. $w = -P_{ext}\,\Delta V = -1 \times (20 - 2) = -18$ bar L. Ans. (2)

Q73. $TV^{\gamma-1} = \text{const}$, $\gamma = \frac{5}{3}$.
$\dfrac{T_2}{T_1} = \left(\dfrac{1}{8}\right)^{2/3} = \dfrac{1}{4}$, to 1 : 4. Ans. (2)

Q74. Adiabatic mein $\Delta U = +w$ (w = work

done on system). "$\Delta U = -w$" galat hai. Ans. (4)

Q75. $C_2H_5Cl + AgF \longrightarrow C_2H_5F + AgCl$: Swarts reaction. Ans. (2)

Q76. Free expansion mein $\Delta U = 0$, $\Delta H = 0$, $w = 0$, lekin $\Delta S > 0$. Isliye $\Delta S = 0$ galat hai. Ans. (2)

Q77. Grignard formation mein reactivity R–I > R–Br > R–Cl. Ans. (1)

Q78. AgCN covalent hai, to N-attack hota hai aur isocyanide (−NC) banta hai. Benzylic Cl hi replace hota hai. Ans. (2)

Q79. Polar protic solvent mein nucleophilicity $I^- > Br^- > Cl^- > F^-$. Statement I galat, II sahi (I^- kam solvated). Ans. (2)

Q80. PhBr $\rightarrow$ PhMgBr $\xrightarrow{CH_3OH}$ benzene (A). Benzene + n-propyl chloride/$AlCl_3$ $\rightarrow$ carbocation rearrangement $\rightarrow$ isopropylbenzene. Ans. (2)

Q81. S_N1 reactivity: p-methoxybenzyl (B) > benzyl (C) > ethyl chloride (A). Increasing: A < C < B. Ans. (2)

Q82. $q = -10$, $w = +25 \Rightarrow \Delta U = q + w = +15$ kJ. Heat system se surroundings mein gayi, aur work system par hua. Options (1), (2), (3) sab galat, to none. Ans. (4)

Q83. Work on system = +ve, work by system = −ve (IUPAC). Dono statements sahi. Ans. (3)

Q84. 3-methylbut-1-ene + HBr $\rightarrow 2^\circ$ carbocation $\rightarrow$ 1,2-H shift $\rightarrow 3^\circ$ carbocation $\rightarrow$ 2-bromo-2-methylbutane. Ans. (4)

Q85. Reversible process mein surroundings hamesha system ke saath equilibrium mein rehte hain. Ans. (3)

Q86. Wurtz reaction: $2\,CH_3CH_2CH_2Cl + 2\,Na \longrightarrow CH_3(CH_2)_4CH_3 + 2\,NaCl$ (n-hexane). Ans. (1)

Q87. Zyada substituted alkene banne wale halide mein elimination tez hota hai. A (isobutylene) > C (propene) > B (ethene). Ans. (4)

Q88. 1-bromo-3-chlorocyclobutane + 2Na (intramolecular Wurtz) $\rightarrow$ bicyclo[1.1.0]butane. Ans. (4)

Q89. S_NAr mein $-NO_2$ (EWG) rate badhate hain. Order: A < B < C. Ans. (4)

Q90. $n = \dfrac{20}{28}$ mol, $\gamma = 1.4$.

$T_2 = T_1 \left(\dfrac{V_1}{V_2}\right)^{\gamma-1} = 300 \times 2^{0.4} \approx$ 396 K. Ans. (2)

BOTANY + ZOOLOGY (Q 91–180) Solutions

Q91. A–III (acute renal failure → kidney transplantation), B–I (ANF, RAAS par check), C–IV (vasopressin → vasoconstriction), D–II (ketonuria → urine mein ketone bodies). Ans. (4)

Q92. Convolvulaceae family order Polymoniales mein aati hai. Ans. (3)

Q93. Animalia–Kingdom, Chordata–Phylum, Dicotyledonae–Class, Angiospermae–Division: A–IV, B–III, C–II, D–I. Ans. (2)

Q94. ICZN = International Code of Zoological Nomenclature. Ans. (3)

Q95. Glomerulus mein ultrafiltration hota hai kyunki protein ke alawa plasma ka sab kuch filter ho jata hai. A aur R dono sahi, R explanation hai. Ans. (2)

Q96. Meselson–Stahl ne semiconservative DNA replication prove kiya. Ans. (3)

Q97. Alag-alag phyla ke sab animals ek hi highest category Kingdom (Animalia) mein aate hain. Ans. (1)

Q98. Eukaryotes mein 3 RNA polymerases hote hain (I, II, III), to Statement I galat. II sahi. Ans. (2)

Q99. Animal ka Phylum = plant ka Division. Ans. (2)

Q100. Lymph ko tissue/interstitial fluid ki tarah bhi kaha jata hai. Baaki options galat hain (colourless hota hai, main circulatory fluid blood hai). Ans. (1)

Q101. Aam aur gehu dono angiosperm hain, to common category Division hai (class alag: dicot/monocot). Ans. (1)

Q102. UTR translate nahi hote (A sahi). Lekin UTRs efficient translation ke liye chahiye, termination ke liye nahi. Termination stop codon karta hai, to R galat. Ans. (1)

Q103. Felidae aur Canidae → order Carnivora. Ans. (3)

Q104. Nucleotide = base + sugar + phosphate. Phosphodiester linkage do nucleotides ke beech hota hai, ek mein nahi. Ans. (2)

Q105. BP aur GFR bahut badhne par ANF blood vessels ko dilate karta hai. Ans. (3)

Q106. Baingan = *Solanum melongena*. Ans. (2)

Q107. B, C, D sahi hain. A galat hai, kyunki DNA directly protein code nahi karta, mRNA ke through karta hai. Ans. (3)

Q108. DNA mein Uracil nahi hota (RNA mein hota hai). Ans. (3)

Q109. A sahi, R sahi aur R, A ki sahi explanation hai. Ans. (3)

Q110. Systemic circulation tissues se CO_2 aur harmful substances door le jata hai. Baaki galat hain: tunica media middle layer hai, QRS = ventricular depolarisation, HR = $\frac{CO}{SV}$. Ans. (3)

Q111. Class mein phylum ke comparison mein zyada similarities hoti hain. Ans. (2)

Q112. Genome size: Human (3.1×10^9 bp) > *E. coli* > lambda > ϕX174. Ans. (3)

Q113. *Triticum aestivum* = wheat. Ans. (2)

Q114. Species se kingdom ki taraf jaane par common characters kam hote jaate hain. Ans. (2)

Q115. Protein synthesis ka factory ribosome hai. Ans. (2)

Q116. Insecta ek class hai, order nahi. Ans. (1)

Q117. Manushya phylum Chordata mein aata hai. Ans. (1)

Q118. Nucleoside = nitrogenous base + pentose sugar (N-glycosidic bond). Ans. (3)

Q119. Dono statements sahi hain. Ans. (3)

Q120. DNA mein base pairing A–T aur G–C. Ans. (2)

Q121. Primata → class Mammalia. Ans. (1)

Q122. Division, Kingdom aur Class ke beech aata hai (plants mein). Ans. (1)

Q123. Order = kai families ka samooh. Ans. (3)

Q124. A sahi (DNA fingerprinting variations pata karta hai). R bhi sahi (uses), lekin R, A ki explanation nahi hai. Ans. (4)

Q125. Avery, MacLeod, McCarty se pehle genetic material protein samjha jata tha. Ans. (3)

Q126. Monkey, gorilla, gibbon sab Mammalia class mein hain. Ans. (3)

Q127. Arthropoda mein housefly aati hai. Ans. (2)

Q128. Division sabse high category hai, to isme similar characters sabse kam hain. Ans. (4)

Q129. Genome ka sirf $< 2\%$ hissa protein code karta hai. Ans. (2)

Q130. Initiator codon–AUG (II), terminator–UAG (IV), phenylalanine codon–UUU (I), initiator tRNA anticodon–UAC (III): A–II, B–IV, C–I, D–III. Ans. (2)

Q131. hnRNA ki capping aur tailing nucleus mein hoti hai. Ans. (3)

Q132. Ab tak known aur described species lagbhag 1.7–1.8 million hain. Ans. (1)

Q133. *pardus*, *leo*, *tigris* teeno *Panthera* mein hain. *indica* alag genus (*Mangifera*) ka species epithet hai. Ans. (3)

Q134. B galat (repeats genome ka bada hissa hain). C galat (chromosome 1 mein sabse zyada genes, Y mein sabse kam). E galat (last sequenced chromosome 1 tha). B, C aur E galat. Ans. (2)

Q135. lac operon ka repressor *i* gene se code hota hai. Ans. (2)

Q136. Tunica intima–squamous endothelium (II), second heart sound–dub (III), monocyte–phagocytic cell (IV), globulin–defense (I): A–II, B–III, C–IV, D–I. Ans. (3)

Q137. Sweat cooling karta hai aur sebum skin ko protective oily covering deta hai. Dono sahi. Ans. (2)

Q138. Digestive tract aur liver ke beech hepatic portal system hai. Ans. (3)

Q139. Har kidney mein lagbhag 1 million nephrons hote hain, billion nahi. Ans. (3)

Q140. II galat (cardiac cycle 0.8 sec). IV galat (lagbhag 80% log Rh^+ hote hain). Ans. (3)

Q141. Wheat $\rightarrow$ family Poaceae. Ans. (2)

Q142. C galat ('O' universal donor hai, 'A' nahi). D galat (haemoglobin RBC mein hota hai, WBC mein nahi). Total 2 galat. Ans. (2)

Q143. Loop of Henle medulla mein bhi jaata hai, to sirf cortex mein nahi hai. Ans. (1)

Q144. Standard ECG: 3 leads, dono wrists aur left ankle par. Ans. (4)

Q145. Okazaki fragments ko jodne wala enzyme DNA ligase hai. Ans. (1)

Q146. Birds ke heart 4 chambered hote hain, 3 chambered galat hai. Ans. (3)

Q147. Collecting duct $\rightarrow$ ureter $\rightarrow$ urinary bladder $\rightarrow$ urethra. Option (4) mein DCT $\rightarrow$ collecting duct se shuru hai, jo sahi order hai. Ans. (4)

Q148. Atrium aur ventricle ke beech thick fibrous atrio-ventricular septum hota hai. Ans. (1)

Q149. Dono statements sahi hain (micturition reflex aur JGA se GFR regulation). Ans. (2)

Q150. Histones positively charged basic proteins hain. Ans. (3)

Q151. Dialysis mein anticoagulant heparin use hota hai. Ans. (1)

Q152. II galat (nodal tissue autoexcitable hai). III galat (heart mesodermal hai). Ans. (2)

Q153. Kidney mein oxalate stones renal calculi kehlate hain. Ans. (2)

Q154. Diastolic pressure 80 mmHg hota hai, 120 mmHg systolic hai. Ans. (4)

Q155. Sahi likhawat: *Mangifera indica* (genus capital, species small). Ans. (1)

Q156. Human heart mein 2 chhote upper chambers (atria) aur 2 bade lower chambers (ventricles) hote hain. Ans. (1)

Q157. Glycosuria–urine mein glucose (II), Malpighian corpuscle–cortex (I), cortical nephrons–reduced vasa recta (IV), reptiles–uricotelic (III). Ans. (3)

Q158. Sabse thick muscles left ventricle mein hote hain (I galat). Left atrium oxygenated blood leta hai (II galat). Ans. (4)

Q159. Glomerular filtrate blood plasma ke isotonic hota hai, lagbhag 300 mOsm L^{-1}. Ans. (3)

Q160. *Panthera leo* = lion. Ans. (1)

Q161. Mammals ureotelic, bony fishes ammonotelic (excretion as ammonia). Ans. (2)

Q162. Thrombocytes–blood clotting, eosinophils–allergic reactions, P-wave–atria ka depolarisation, universal recipient–AB group: A–II, B–III, C–IV, D–I. Ans. (3)

Q163. ADH (antidiuretic hormone) diuresis rokta hai. Ans. (2)

Q164. Lymphocyte agranulocyte hai. Ans. (3)

Q165. Land insects aur snails uric acid nikalte hain (A sahi), aur R sahi explanation hai (water conservation). Ans. (3)

Q166. SAN: right atrium ka right upper corner. AVN: right atrium ka lower left corner. Ans. (2)

Q167. ADH neurohypophysis se aata hai (A galat). Renin kidney (JG cells) se aata hai (D galat). Ans. (4)

Q168. QRS complex ventricular contraction (depolarisation) dikhata hai. Ans. (2)

Q169. Nerves aur blood vessels kidney mein hilum se enter karte hain. Ans. (3)

Q170. Heart sounds sunne ke liye stethoscope use hota hai. Ans. (1)

Q171. Podocytes Bowman's capsule ki wall mein hote hain. Ans. (3)

Q172. Globulin antibodies banate hain, to iski kami se body ka defense mechanism affect hoga. Ans. (2)

Q173. Invertebrates mein simple tubular forms hote hain (protonephridia, nephridia etc.), aur vertebrates mein complex tubular organs (kidney). Dono statements ulte hain, isliye dono galat. Ans. (4)

Q174. A sahi (myogenic), C sahi (CAD = atherosclerosis). B galat, kyunki medulla ka centre ANS ke through control karta hai, CNS ke through nahi. D galat, kyunki parasympathetic signals cardiac output ghatate hain. A aur C sahi. Ans. (4)

Q175. Diagram mein label A outer region ko point kar raha hai, jo cortex hai. *(Note: figure se judge kiya gaya hai, ek baar original paper se verify kar lena.)* Ans. (2)

Q176. Emergency mein O aur Rh negative (universal donor) safe hai. Ans. (2)

Q177. Platyhelminthes mein osmoregulation aur excretion flame cells karte hain. Ans. (2)

Q178. Open circulation molluscs aur cockroach mein hota hai. Ans. (4)

Q179. Uric acid pellet/paste ke form mein land insects aur reptiles nikalte hain. Ans. (3)

Q180. Mitral valve–left atrium aur left ventricle ke beech ke opening ko guard karta hai (II). Neutrophils–phagocytic action (IV). Basophils–inflammatory reactions (I). Erythrocytes–gases ka transport (III). Ans. (3)